

Modules over PID's

Recall:

$$(3) \quad R = k[x]$$

• An isomorphism of k -pairs

$$(V_1, T_1) \xrightarrow{\sim} (V_2, T_2)$$

is a k -linear isomorphism

$$\varphi: V_1 \longrightarrow V_2$$

s.t.

$$\begin{array}{ccc} V_1 & \xrightarrow{\varphi} & V_2 \\ T_1 \downarrow & & \downarrow T_2 \\ V_1 & \xrightarrow{\varphi} & V_2 \end{array}$$

$$T_2 \circ \varphi = \varphi \circ T_1$$

$$\Updownarrow$$

$$T_2 = \varphi \circ T_1 \circ \varphi^{-1}$$

$$\left\{ \begin{array}{l} k[x] \text{-linear} \\ \text{isomorphisms} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{isomorphisms} \\ \text{of } k\text{-pairs} \end{array} \right\}$$

Pf:

$$(V_1, T_1) \quad (V_2, T_2) :$$

$$\left\{ \begin{array}{l} k[x]\text{-module } V_1 \\ p(x) \cdot v = p(T_1)(v) \end{array} \right.$$

$$\left\{ \begin{array}{l} k[x]\text{-module } V_2 \\ p(x) \cdot v = p(T_2)(v) \end{array} \right.$$

$$(V_1, T_1) \xrightarrow{\cong} (V_2, T_2)$$

isomorphism
of k -pairs

$$V_1 \xrightarrow[\varphi]{\cong} V_2 : \text{isomorphism of } k\text{-vector spaces}$$

$$\text{s.t. } T_2 = \varphi \circ T_1 \circ \varphi^{-1}$$

Claim: φ is a $k[x]$ -linear isomorphism

Pf: $\varphi(p(x) \cdot v_i) = p(x) \cdot \varphi(v_i)$

$$p(x) = a_n x^n + \dots + a_0$$

||

$$\forall p(x) \in k[x], v_i \in V_i$$

$$\varphi([a_n x^n + \dots + a_0] \cdot v_i)$$

$$[a_n x^n + \dots + a_0] \cdot \varphi(v_i)$$

||

||

$$\varphi(a_n x^n \cdot v_i + \dots + a_0 \cdot v_i)$$

|| φ is k -linear

$$a_n x^n \cdot \varphi(v_i) + \dots + a_1 x \cdot \varphi(v_i) + a_0 \cdot \varphi(v_i)$$

$$a_n \varphi(x^n \cdot v_i) + \dots + a_1 \varphi(x \cdot v_i) + a_0 \varphi(v_i)$$

||
?
?

To see this, it is enough to know

that

$$\varphi(T_1(v_i)) = T_2(\varphi(v_i)) \iff$$

$$\varphi(x \cdot v_i) = x \cdot \varphi(v_i)$$

$$\forall v_i \in V_i$$

||

$$\varphi \circ T_1 = T_2 \circ \varphi$$

||

$$\varphi \circ T_1 \circ \varphi^{-1} = T_2$$

Upshot:

Classifying $A \in M_n(F)$ up
to conjugacy

HW 6

($\Rightarrow$) Classifying n -dim'l
problem 2 k -pairs up to
isomorphism

($\Leftarrow$) Classifying $k[x]$ -modules
up to isomorphism where
the underlying k -vector space
is n -dimensional

Defn A commutative ring
 R is a principal ideal domain
or PID if it is an
integral domain
if every ideal in R is principal

Examples

(1) $k[x]$, k : field

(2) Any field k . Ignore.

(3) $\mathbb{Z}$

(4) $\mathbb{Z}[i] = \{a+bi: a, b \in \mathbb{Z}\}$
 $\leq \mathbb{C}$

(Gaussian integers)

Defⁿ (1) $r \in R$ is irreducible

if it is not invertible [i.e. $\nexists s \in R$ s.t. $rs=1$]

and if r cannot be written as
a product $r = s_1 s_2$ of two
non-invertible elements $s_1, s_2 \in R$.

Note: This is only meaningful if
 R is not a field.

(2) $r \in R$ is prime if

(r) is a prime ideal

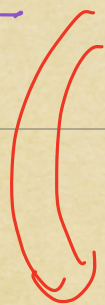
Thms $(\Leftrightarrow)$ If $ab \in (r)$ then
either $a \in (r)$ or $b \in (r)$

$(\Leftrightarrow) R/(r)$ is an

• integral domain

Proposition: $R: \text{PID}$

TFAE: (1) r is irreducible



(2) r is prime

$\Uparrow$ general fact: maximal $\Rightarrow$ prime

(3) (r) is maximal

$R/(r)$ is a field

Recall

m
 $\Uparrow$
 $\mathbb{Z}$

$f(x)$

$\Uparrow$
 $k[x]$

• m is prime

• $\mathbb{Z}/m\mathbb{Z}$ is an integral domain

• $\mathbb{Z}/m\mathbb{Z}$ is a field

• $f(x)$ is irreducible

• $k[x]/(f(x))$ is an integral domain

• $k[x]/(f(x))$ is a field

Pf of Proposition:

$$(1) \Rightarrow (3)$$

If (r) is not maximal, $\exists$ ideal I

$$(r) \subsetneq I \subsetneq R$$

But $I = (s)$, $s \in R$. [because R is a PID]

$$(r) \subsetneq (s)$$

$$\Rightarrow r = st, \quad \exists t \in R$$

$\uparrow$ non-invertible because $(s) \neq R$
 $\nwarrow$ non-invertible because $(s) \neq (s)$

This shows r is reducible

not (3) $\Rightarrow$ not (1) (Contrapositive)

$$(2) \Rightarrow (1)$$